AI-Enabled Smart Electric Vehicle Charging and Energy Management Platform

Sub-Project 1 —Phase 1 Report

Data acquisition, preprocessing and demand-forecasting baseline

Sub-project	SP1 —EV charging data analysis and demand forecasting
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Date	14 September 2026
Deliverables covered	Clean datasets; analysis framework; forecasting models; accuracy evaluation
Code	`subprojects/sp1-data-forecasting/`
Outputs	`data/processed/sp1-demand-forecast-v1.csv`
	`data/processed/sp1-baseline-demand-v1.csv`

1 Scope and objectives

SP1 owns the first stage of the platform's data flow: it turns raw public charging records into the gap-free hourly demand series that every other sub-project consumes (SP2 schedules against it, SP4 displays it, SP5 prices it). The official brief assigns SP1 four tasks: collect and preprocess public EV charging datasets; analyse charging patterns and user behaviour; develop machine-learning models to predict future charging demand; and evaluate forecasting accuracy and model performance. This report covers all four, and with them the Phase-1 milestone (datasets secured, tools aligned, framework running).

Two constraints shaped the work and are resolved in Section 2:

1. **Dataset selection.** The supervisor's kickoff email named three sources —Caltech ACN-Data, Boulder Colorado, and UK National Grid —and each was checked against its live service. The outcome decided the project's data foundation.
2. **Official provenance.** The earlier plan assumed Chinese datasets. A member-account review of the China Charging Alliance platform and a search of provincial and grid-operator portals found no session-level Chinese data, so that route could not carry a forecasting model.

2 Data acquisition and source evaluation

2.1 Sources verified

Every source below was probed with live requests (HTTP status, schema, row counts, licence text); nothing is quoted from a landing page without having been observed. The full verification log is kept in

the repository at `data/raw/official-datasets-research.md`.

Source	Official?	EV charging data?	Access observed	Verdict
City of Boulder, Colorado	Yes — municipal government	Yes — sessions, metered kWh	ArcGIS FeatureServer, no registration; 148,136 rows pulled	**In use (primary)** CC0 1.0
City of Palo Alto, California	Yes — municipal government	Yes — sessions, metered kWh, user IDs	Direct CSV, no registration; 85.4 MB, 259,415 rows	**In use (validation)** PDDL
Caltech ACN-Data	Yes —Caltech + PowerFlex	Yes — sessions *plus* measured power time series	API returns HTTP 401; free self-registration issues a token	Obtainable; not used yet (see §6)
UK National Grid (now NESO)	Yes —national system operator	**No** —GB national demand only	CKAN/CSV, no registration, 2001–2025	Not an EV dataset; grid context only
China Charging Alliance	Yes — industry alliance	**No** — monthly aggregates only	Member account required	Could not support SP1

Table 1: Sources checked for SP1. Only the first two provide session-level EV charging records that are official *and* downloadable without a signed agreement.

Two consequences follow, and both are recorded in the integration contract as amendment A1 rather than being decided quietly:

- **Region.** The project can no longer be described as China-only; the demand series is for two US cities. The file formats are unchanged.
- **Currency.** Neither dataset contains a tariff or price column, so CNY pricing has no basis. SP2 must source a US tariff schedule before it can populate `tariff_cny_per_kwh`; that is now an open item owned by SP2.

2.2 Datasets used

Property	City of Boulder (primary)	City of Palo Alto (validation)
Publisher	City of Boulder Open Data	City of Palo Alto Open Data
Licence	CC0 1.0 (public domain)	PDDL (public domain)
Sessions	148,136	259,415
Stations	50 (all Level 2)	47
Period	2018-01-02 – 2023-12-01 (5.9 years)	2011-07-30 – 2021-01-01 (9.4 years)
Energy column	`Energy (kWh)` —**metered**	`Energy (kWh)` —**metered**
Timestamps	start and end, minute resolution	start and end, minute resolution
Per-vehicle ID	none	`User ID`, 97% populated
Hourly series	*derived* (see §3)	*derived*
Rows with zero energy	16,113 (10.9%)	—

Table 2: The two datasets actually used. Both are ChargePoint-style exports, so both share the same traps, handled below.

2.3 Cleaning decisions

The datasets are published as session records, so "cleaning" here means mapping them into one canonical schema and refusing to let known traps through silently.

1. **Occupancy and charging time.** Both sources publish a total plugged-in duration *and* an active charging duration. In Boulder the median occupancy is 1.55 h but the tail reaches 839 h (vehicles left connected for weeks). Spreading a session's energy across occupancy would smear the load profile into nonsense, so the adapter maps the canonical duration to *charging* time (median 1.39 h, 99th percentile 8.3 h) and defines the session window as `start` → `start + charging_time`.

2. **Units and types.** Boulder delivers every column as a string, including numerics; the adapter casts explicitly and reports unparsable values instead of dropping rows. Durations arrive as `hh:mm:ss` rather than minutes, which the adapter now parses in either form.

3. **Timezone.** Storage is UTC. Source timestamps are naive local time; each dataset declares its own zone (`America/Denver`, `America/Los_Angeles`) and is converted on load, so daylight saving is handled by the timezone database rather than by the file's own (sometimes inconsistent) zone label.

4. **Energy provenance.** These datasets measure delivered energy. Where a dataset instead publishes only a *rated* power, energy becomes a derived upper bound; that is a materially weaker input and was one reason the Chinese third-party dataset was set aside.

3 Method

3.1 From sessions to an hourly demand series

Each session contributes its metered energy to every hourly interval it overlaps, in proportion to the overlap duration. The result is energy per interval, which at a one-hour resolution is numerically the average power in kW. Absent intervals mean "no charging recorded", which for demand is a true zero, so the grid is completed with zeros; the contract's guarantee to downstream sub-projects is a *gap-free, non-negative, UTC-stamped* series, and the pipeline re-reads and validates every file it writes.

One modelling assumption applies throughout: the datasets contain no intra-session power trace, so a session's energy is spread *evenly* across its charging window. Real charging curves taper, so this slightly smooths peaks within a session. The assumption is recorded in each output's `.meta.json` under `interval_method`, so no downstream sub-project has to infer it.

3.2 The uncontrolled baseline

SP2 and SP5 need something to compare an optimised schedule against, so SP1 also emits an "uncontrolled" profile: every session's energy is moved to a block starting at 18:00 local on the day it occurred, charged at that session's own recorded power. Energy is therefore conserved and only its timing changes —a deliberate worst-case shape, not a network simulation. (At a fixed 7 kW this would stretch a 150 kW fast-charge session over 40 hours; using each session's own power avoids that artefact.)

3.3 Features and forecasting

Forecasting is framed as *direct multi-horizon forecasting with the horizon as a feature*: one model predicts all 24 hours from a single origin, so errors do not accumulate the way they do in recursive one-step prediction.

- **Known-future features** describe the *target* interval: local hour of day, day of week, weekend flag, month, and cyclical encodings. These are known in advance, so using them is legitimate.
- **Autoregressive features** describe the *origin*: demand lags of 1, 2, 3, 24, 48 and 168 hours and rolling means, standard deviations and maxima over 24 and 168 hours, all shifted so they cannot see the interval being predicted.
- **Naive references** are first-class inputs: the value 24 hours and 168 hours before the target, which also serve as the persistence baselines.

A regression test recomputes every lag from the raw series and fails if any feature can see its own target; a second one pins the training boundary with a ramp series, because a leaky backtest reports fantasy accuracy.

3.4 Models and evaluation

Seven models are scored: three baselines (`naive_daily`, `naive_weekly`, `hour_of_week_profile`) and four learned models (ridge regression, random forest, gradient boosting, XGBoost). Evaluation uses a rolling-origin backtest: for each origin the model is retrained using only targets strictly before that origin, and forecasts the next 24 hours. Origins are anchored to local midnight, so each is a genuine day-ahead forecast, and only origins with a full horizon are used, which keeps the evaluation window gap-free. Training history per origin is capped at 180 days.

Accuracy is reported as WAPE (total absolute error divided by total demand), which stays stable when many intervals have near-zero demand, alongside MAE, RMSE and R^2. Prediction intervals are empirical: $\hat{y} \pm 1.645\,\sigma$ where σ is the residual standard deviation at the same horizon in the backtest, clipped at zero.

4 Results

4.1 Charging patterns

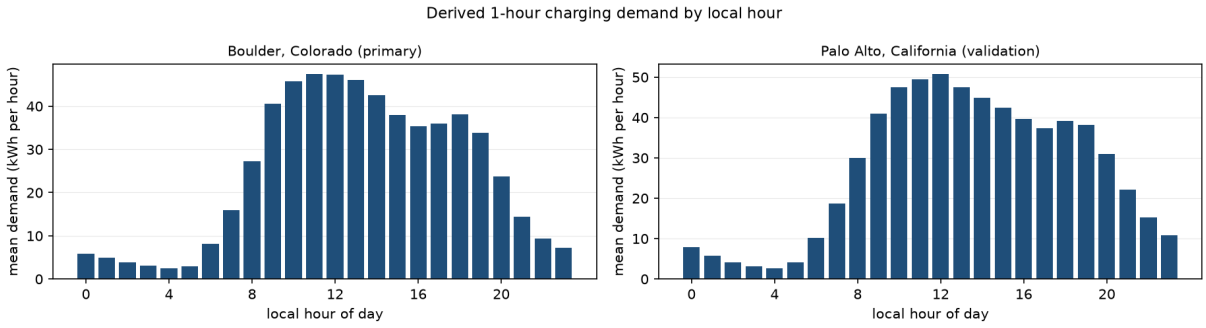


Figure 1: Mean derived demand by local hour of day. Both sites are Level-2 workplace and municipal charging, so both peak in the middle of the working day rather than in the evening —the residential 18:00 peak of the project brief does not appear in either dataset.

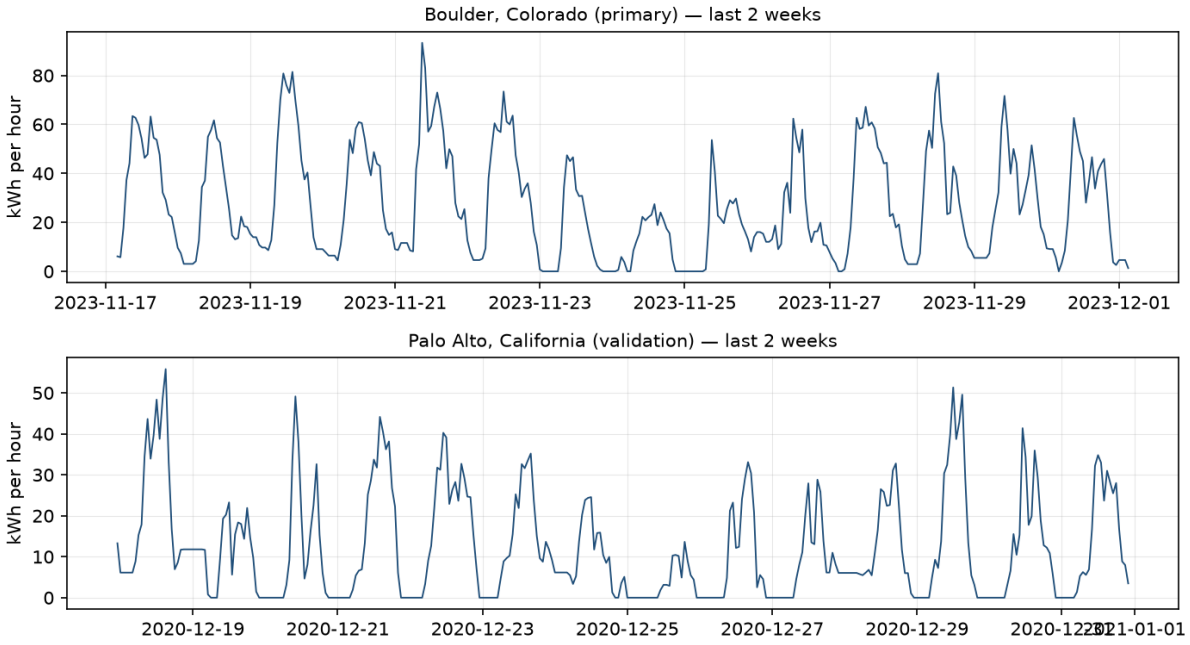


Figure 2: The last two weeks of each hourly series, in local time: a strong daily cycle with visibly lower weekend activity.

Boulder's 148,136 sessions aggregate to 1.253 GWh over 5.9 years. The local-hour profile (Figure 1) has the shape of a workplace fleet: demand climbs from 06:00, peaks in the late morning (11:00 carries 8.2% of daily energy), and decays after 20:00 to a 04:00 trough (0.42%). The peak-to-trough ratio is 19.3×, so the *timing* of charging dominates even at this small scale. A secondary bump at 17:00–18:00 (6.2–6.6%) is consistent with vehicles plugging in at the end of the working day. Palo Alto's 259,415 sessions give 2.216 GWh over 9.4 years with a similar shape: peak at 12:00 (7.9%) and a 04:00 trough (0.41%), peak-to-trough 19.2×. Two independently published municipal networks showing the same daily signature supports the preprocessing: the shape is a property of Level-2 workplace charging rather than of one city. It also means the brief's evening residential peak is absent from both datasets, which the Phase-3 scenario design has to acknowledge rather than assume.

4.2 Forecast accuracy

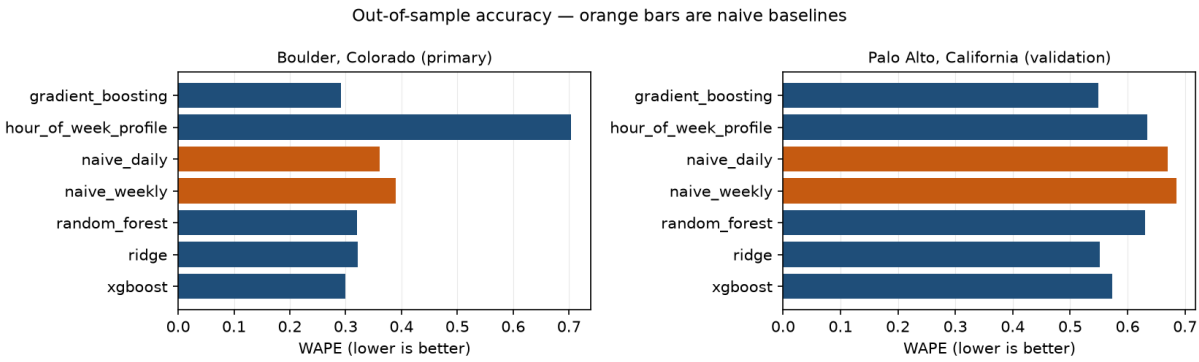


Figure 3: Out-of-sample WAPE by model. Orange bars mark the naive baselines used as the comparison floor.

Boulder contributes 25 daily origins, that is 600 out-of-sample hours. Gradient boosting attains the lowest WAPE, every learned model outperforms both naive baselines, and the ranking holds across the horizon, with per-horizon WAPE between 0.196 and 0.910.

Model	WAPE	MAE (kWh/h)	RMSE (kWh/h)	R^2
gradient boosting	**0.2920**	8.37	11.39	0.763
XGBoost	0.2999	8.60	11.79	0.747
random forest	0.3197	9.17	12.53	0.714
ridge regression	0.3217	9.23	12.25	0.727
naive daily	0.3602	10.33	13.97	0.644
naive weekly	0.3891	11.16	15.45	0.565
hour-of-week profile	0.7040	20.19	25.76	−0.210

Table 3: Boulder (primary): Out-of-sample accuracy on the primary site. 25 daily origins, 600 out-of-sample intervals, all models retrained per origin.

Gradient boosting reduces relative WAPE by 19% against daily persistence (0.292 against 0.360). The seasonal-profile baseline is the only model with a negative R^2, since averaging by weekday and hour cannot follow the drift visible in Figure 2.

Palo Alto contributes 14 daily origins, 336 out-of-sample hours. The same model attains the lowest WAPE and the learned models again outperform persistence (0.549 against 0.670), but absolute accuracy is markedly lower. Its evaluation window ends on 2020-12-31, inside the COVID-19 period when workplace commuting collapsed, which accounts for the difference.

Model	WAPE	MAE (kWh/h)	RMSE (kWh/h)	R^2
gradient boosting	**0.5491**	7.02	9.70	0.466
ridge regression	0.5514	7.05	9.79	0.456
XGBoost	0.5731	7.33	9.98	0.434
random forest	0.6301	8.06	10.84	0.333
hour-of-week profile	0.6341	8.11	11.17	0.292
naive daily	0.6702	8.57	11.74	0.217
naive weekly	0.6846	8.75	12.63	0.094

Table 4: Palo Alto (validation): Out-of-sample accuracy on the validation site. 14 daily origins, 336 out-of-sample intervals, all models retrained per origin.

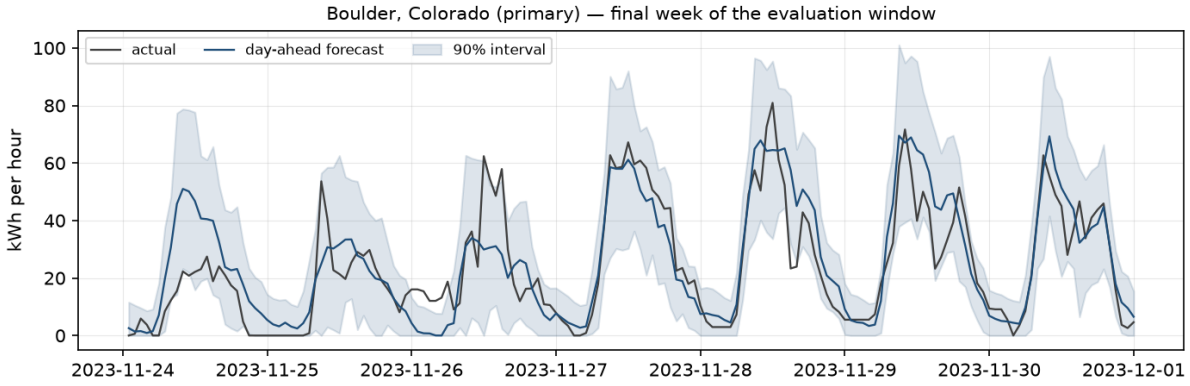


Figure 4: Final week of the Boulder evaluation window: the day-ahead forecast against the realised demand, with the empirical 90% interval.

4.3 Uncontrolled baseline versus forecast

The second contract file moves every session to an 18:00 local start at that session's own power, which conserves energy and changes only its timing. Over Boulder's 600-hour window it raises the peak from 94.5 kWh to 362.5 kWh, an amplification of $3.8\times$, while total energy is preserved to within $+0.05\%$ of the realised demand. The brief's "50 EVs plug in at 18:00" scenario corresponds to that amplification, and it is the quantity SP2's scheduler is intended to reduce.

Quantity (600 h window)	Day-ahead forecast	Uncontrolled baseline
Total energy	17.81 MWh	17.22 MWh (realised 17.21 MWh)
Peak hour	94.5 kWh	**362.5** kWh
Mean load	29.7 kWh/h	28.7 kWh/h
Energy bias against realised demand	+3.5%	+0.05%
90% interval, mean width	33.2 kWh	not modelled

Table 5: Boulder: the two contract deliverables over the same window. The uncontrolled profile moves every session to an 18:00 local start at its own recorded power, so energy is conserved and only its timing changes.

The empirical coverage of the 90% interval is 91.0% on Boulder and 90.5% on Palo Alto, so the residual-based interval is calibrated rather than nominal. Its mean width, 33.2 kWh, exceeds the mean demand, which reflects how spiky the series is at this scale.

5 Limitations and threats to validity

1. **Derived hourly curve.** Both datasets are session-level; no native hourly or sub-hourly power series exists in either. The derivation is documented and its plausibility was checked (Palo Alto's implied average power has a median of 3.7 kW, consistent with Level-2 AC charging), but a within-session taper is not represented.

2. **Fleet size and charging level.** Boulder has 50 stations, Palo Alto 47, and both are workplace/municipal AC charging. The brief's "50 EVs plug in at 18:00" scenario is therefore constructed as a stated scenario rather than read off the data; the grid-stress question is about *when* energy is drawn, which these datasets do support.

3. **Region.** The original framing was a Chinese market project; the data are US municipal. Market relevance was traded for data authenticity and licensing, and the supervisor accepts either region.

4. **Tariff data.** Neither dataset carries a price column, so SP1 cannot produce cost; SP2 must source a US tariff schedule.

5. **External shocks.** Palo Alto's record runs to the end of 2020, so its final evaluation window sits in the COVID-19 period, when commuting and therefore workplace charging collapsed. Comparisons between the two sites have to account for that.

6. **Evaluation sample.** A few dozen daily origins per site at a 24-hour horizon give a stable but not large sample; the ranking should be re-checked as more history accumulates.

6 Phase-2 plan

1. **Forecast granularity.** Palo Alto carries a user identifier for 97% of sessions, so a per-EV forecast is a scope decision rather than a data limitation. It is the interface question that still constrains SP2's optimiser design.
2. **Third site.** Caltech ACN-Data is the only source reviewed that publishes measured power time series, which would allow the cost of the even-spreading assumption to be measured. It requires a free research registration.
3. **Tariff schedule.** Sourcing a US tariff (open item for SP2) is what unblocks the economic chain SP2 $\rightarrow$ SP5.
4. **Model comparison.** Longer training windows, a per-site model, and an explicit test of whether the learned models' advantage over persistence survives at longer horizons.
5. **Literature review.** The SP1-relevant strands (demand forecasting for EV charging, and session-to-load disaggregation) are covered in the companion review document.